MIL-T-10794G
20 April 1982
SUPERSEDING
MIL-T-10794F
1 April 1977

MILITARY SPECIFICATION

TUBES, ALUMINUM-ALLOY, EXTRUDED PIPELINE SECTION, WITH GROOVED NIPPLE WELDED ON EACH END

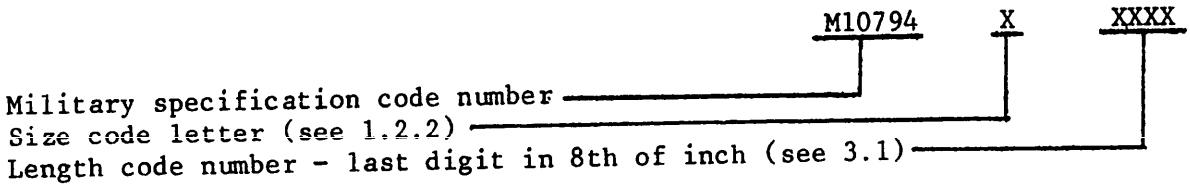
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pipeline section consisting of an extruded aluminum-alloy thin-wall tube with an extruded aluminum-alloy thick-wall grooved-nipple welded on each end.

1.2 Classification. Tubing shall be the following sizes, and shall be designated by the applicable definitive military specification part number (see 6.2 and 6.6).

1.2.1 Definitive military specification part number. Tubing covered by this specification shall be designated in the following form (see 6.2):



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4710

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1.2.2 Size. The tubing size is identified by a single letter (see table I).

TABLE I. Size code letter.

Size code Letter	Nominal size (inches)
A	4
B	6
C	8
D	12

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-C-10387 - Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe and Tube.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM B 345 - Aluminum-Alloy Seamless Extruded Tube and Seamless Pipe for Gas and Oil Transmission and Distribution Piping Systems.

ANSI B46.1 - Surface Texture (Surface Roughness, Waviness and Lay).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

A5.10 - Specification for Aluminum and Aluminum Alloy Bare Welding Rods and Electrodes.

D10.9 - Specification for Qualification of Welding Procedures and Welders for Piping and Tubing.

(Application for copies should be addressed to the American Welding Society Inc., 2501 North West Seventh Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

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(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 Description. The pipeline sections shall be as shown on figure 1 and as specified herein. The pipeline section shall consist of a light-gage tube and two grooved nipples. The nipples shall be connected to the ends of the light-gage tube with a circumferential weld and shall be compatible with couplings conforming to MIL-C-10387. Unless otherwise specified (see 1.2.1), the length of the completed pipeline section shall be 240 inches, plus or minus 1/4 inch.

3.2 First article (preproduction model). The contractor shall furnish one or more pipeline sections as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce pipeline sections that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Aluminum filler metal shall conform to AWS A5.10, type ER 5356 or ER 5556 (see 6.5).

3.4 Light-gage tube. The light-gage tube shall be an extrusion of aluminum alloy conforming to ANSI/ASTM B 345, UNS A96061, temper T6. The tube shall be continuous within its specified length.

3.5 Nipples. Nipples shall be fabricated from extrusions of a aluminum-alloy conforming to ANSI/ASTM B 345, UNS A96061, temper T6. The land-sealing surface on each nipple shall be smooth, free from burrs, scratches, or other indentations and shall have a roughness height rating of not more than 125 as specified in ANSI B46.1. Machining of the nipple shall be done in such a manner that the wall thickness under the coupling groove shall be not less than the minimum allowable light-gage tube wall thickness.

3.6 Pressure ratings. The tubing and nipple shall not leak at the welded joint and shall withstand the test pressures shown in table II.

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TABLE II. Pressure ratings.

Size	Working pressure	Hydrostatic test pressure	Minimum hydrostatic burst pressure
(inches)	(psi)	(psi)	(psi)
4	600	900	1200
6	600	900	1200
8	500	750	1000
12	400	600	800

3.7 Identification marking. Each pipeline section shall be identified in accordance with MIL-STD-130. Marking shall be placed on one end nipple not less than 1 inch from the circumferential weld.

3.8 Treatment and painting. Unless otherwise specified (see 6.2), all exterior surfaces of the tubing shall be treated and painted in accordance with MIL-T-704 type A, color forest green. The interior surfaces of each pipeline section shall be thoroughly cleaned of all dirt and foreign matter.

3.9 Workmanship.

3.9.1 Welders. Before assigning any welder to manual or automatic welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed:

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
AWS D10.9, Specification for Qualification of Welding Procedures and Welders for Piping and Tubing.

3.9.2 Welding. The surface of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. All welding shall be done by the inert-gas shielded-metal-arc welding process without the use of flux. In butt-welded joints, weld beads shall extend not more than 3/32 inch into the inner periphery of the pipeline section (see figure 1) and shall be free of sharp projections.

3.9.2.1 Repair of defective welds. Minor defects such as sweats or leaks in the welds may be repaired. Prior to repair welding, the defective area shall be completely dried using flame or hot air, but no part of the tube section shall be heated to more than 399° F.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsiblility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction pipeline section shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction pipeline section shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection lot. All units of the same size, offered to the Government at one time, shall be considered a lot for purposes of inspection.

4.4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.3 Examination.

4.4.3.1 Samples. Samples selected in accordance with 4.4.2 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.4 Tests.

4.4.4.1 Individual. Each pipeline section shall be tested as specified in 4.5.2.1. Failure of this test shall be cause for rejection.

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4.4.4.2 Burst test sampling. Sampling for burst test (see 4.5.2.2) shall be in accordance with MIL-STD-105, inspection level S-2, AQL shall be 4.0 percent defective. Rejected lots shall be scrapped. No item from a rejected lot shall be reworked, screened, or otherwise resubmitted for Government acceptance. Individual sections subjected to the burst test shall not be included in any shipment of acceptable pipeline sections.

4.5 Inspection procedure.

4.5.1 Examination. The pipeline sections shall be examined as specified herein for the following defects:

101. Dimensions not as specified.
102. Diameter and wall thickness tolerance not as specified.
103. Tube length not as specified.
104. Material not as specified.
105. Nipple out of alinement with tubing.
106. Identification marking missing, incomplete, or illegible.
107. Treatment and painting not as specified.
108. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Hydrostatic pressure test. Each pipeline section shall be subjected to a hydrostatic pressure test at the applicable pressure specified in table II. The pressure shall be applied for not less than 30 seconds, and the pipeline section shall then be checked to assure that there are no leaks at the welded joints. The test shall be performed with the ends of the pipeline section attached to couplings with a blank end conforming to MIL-C-10387. The longitudinal expansion of the pipeline section shall not be restricted while under test. While under the specified pressure, each section shall be struck one blow on the circumferential weld with a force equivalent to the impact delivered by a 2-pound steel ball dropped a vertical distance of 5 feet. Evidence of leaking or yielding shall constitute failure of this test.

4.5.2.2 Hydrostatic burst test. The test shall be performed using couplings with a blank end as specified in 4.5.2.1. The longitudinal expansion of the pipeline section shall not be restricted while under test. Evidence of bursting at a pressure less than the applicable burst pressure shown in table II shall constitute failure of this test.

4.5.2.3 Test safety. Hydrostatic pressure and burst tests shall be performed at an isolated or guarded area to prevent potential injury to personnel in the event of material or weld failure. All impact tests will be done remotely.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

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4.6.1.1 Unit of product. For the purpose of inspection, a completed bundle of loose pipeline sections prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109.	Materials not as specified.	Each incorrect material shall constitute one defect.
110.	Land-sealing surface not completely covered with barrier material and tape as specified.	
111.	Application of barrier material and tape not in accordance with figure 2 as specified.	
112.	Tape not rolled in place on the tube and on each end wall.	
113.	Bundling of 4- and 6-inch pipeline sections not as specified.	
114.	Strapping not as specified.	
115.	Marking illegible, incomplete, incorrect, or missing.	

5. PACKAGING

5.1 Preservation. The land-sealing surface of the tube shall be completely covered with a single wrap of barrier material conforming to MIL-B-121, type II, class 2 (see figure 2). The wrap for the land-sealing surface shall be flush with the end of the tube and shall extend back beyond the groove not less than 1/4 inch. The wrap of the barrier material shall be covered with a single wrap of tape conforming to PPP-T-60, type IV. The tape shall be of a width that will completely cover the barrier material and shall extend not less than 1/8 inch beyond the end of the tube and extend back beyond the barrier material not less than 1/2 inch. The end of the tape extending beyond the end of the tube, and the tape extending back beyond the wrap, shall be rolled in place against the end wall of the tube and against the tube (see figure 2).

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pipeline sections (4- and 6-inch nominal sizes). The pipeline sections shall be bundled seven pieces to a bundle. Each bundle shall be secured with four flat steel straps conforming to QQ-S-781, type I, finish B, size 5/8 by 0.023 inch. The two end straps shall be placed approximately 18 inches from the ends and the two intermediate straps shall be placed equidistantly between the end straps.

5.2.1.2 Pipeline sections (8- and 12-inch nominal sizes). The pipeline sections shall be shipped loose.

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5.2.2 Level B. Unless otherwise specified (see 6.2), the pipeline sections shall be shipped loose. When specified (see 6.2), 4- and 6-inch nominal size pipeline sections shall be bundled as specified for level A, except that straps shall not be required to be finish B.

5.2.3 Level C. The pipeline sections shall be bundled or shipped loose as applicable to size, in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Bundles and loose pieces shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The pipeline sections, connected by grooved-pipe-type couplings, are primarily intended for use as military pipelines for conveying liquid petroleum products.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Definitive military specification part number required (see 1.2).

c. Time frame required for submission of the preproduction model(s) and number of pipeline sections required (see 3.2).

d. When treatment and painting for exterior surfaces of the tubing is not required (see 3.8).

e. Degree of packing required (see 5.2).

f. When bundling of 4- and 6-inch pipeline sections is required for level B (see 5.2.2).

6.3 Preproduction model. Any changes or deviations of production pipeline sections from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish pipeline sections conforming to this specification.

6.4 International standardization agreement. Certain provisions of this specification (dimensions) are the subject of international standardization agreement (NATO STANAG No. 2761 and QSTAG-240). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.6 Definitive military specification part number. The military specification part number is a definitive part number which corresponds to the size and length of tubing covered by this specification. The military specification number, size code letters and length code number are combined to form the definitive military specification part number.

6.7 Materiel deterioration prevention and control (MADPAC). The contracting officer should consider for inclusion in all applicable invitations for bid, requests for proposal, and contracts the following statement:

"The technical proposal shall provide assurance that all system/item parts shall be fabricated from compatible materials, suitably finished, to provide protection from corrosion and other forms of materiel deterioration. The assurance shall be in the form of a finish plan, utilizing state-of-art technology, which clearly identifies the specific finish or treatment to be used on the various components and assemblies, to provide protection against the various forms of materiel deterioration in the applicable operating and storage environments to which they may be exposed."

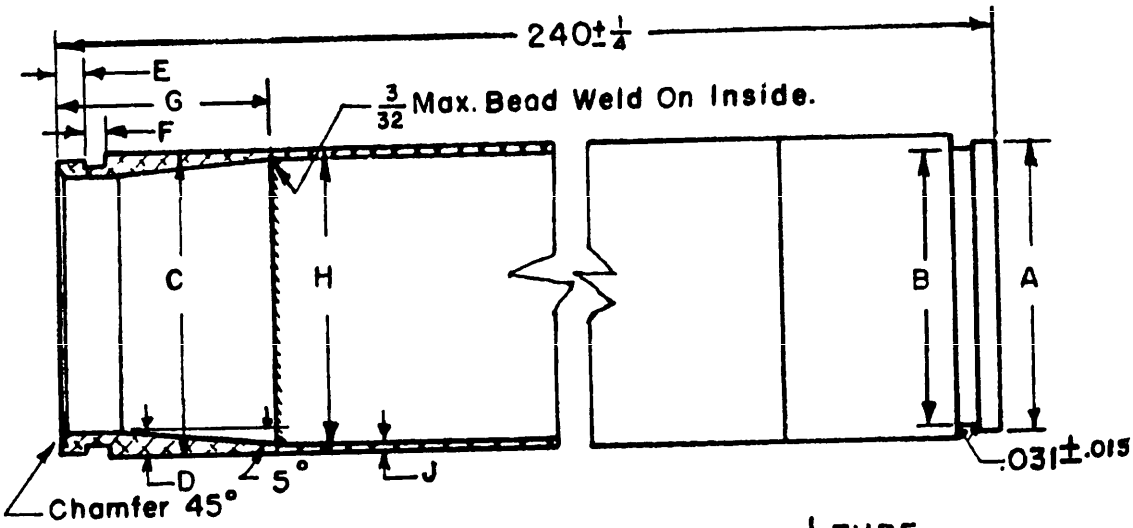
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Preparing activity:
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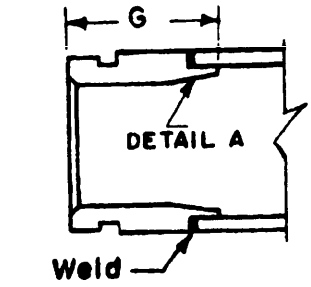
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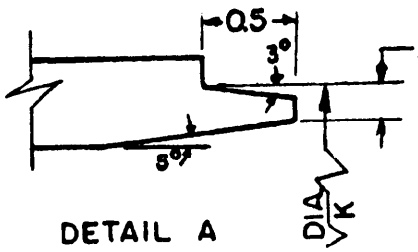
	NIPPLE							TUBE		
SIZE	A	B	C	D	E	F	G	H	J	MIN. J (ECCENTRICITY)
4	4.532 +0 −.016	4.334 +0 −.020	4.566 ±.025	.268 ±.016	5/8 ± 1/32	3/8 ± 1/32	4 ± 1/16	4.566 ±.025	.094 ±.010	.074
6	6.657 +0 −.016	6.445 +0 −.022	6.716 ±.035	.345 ±.025	5/8 ± 1/32	3/8 ± 1/32	6 ± 1/16	6.716 ±.035	.140 ±.020	.108
8	8.657 +0 −.016	8.441 +0 −.025	8.741 ±.045	.402 ±.035	3/4 ± 1/32	7/16 ± 1/32	8 ± 1/16	8.741 ±.045	.150 ±.020	.117
12	12.797 +0 −.032	12.531 +0 −.030	12.915 ±.065	.515 ±.045	3/4 ± 1/32	1/2 ± 1/32	8 ± 1/16	12.915 ±.065	.175 ±.020	.140

OPTIONAL METHOD



All Nipple Dimensions Same As Above Except As Noted On Detail A

SIZE	K
4	4.346 +0 −.008
6	6.380 +0 −.016
8	8.375 +0 −.016
12	12.500 +0 −.016



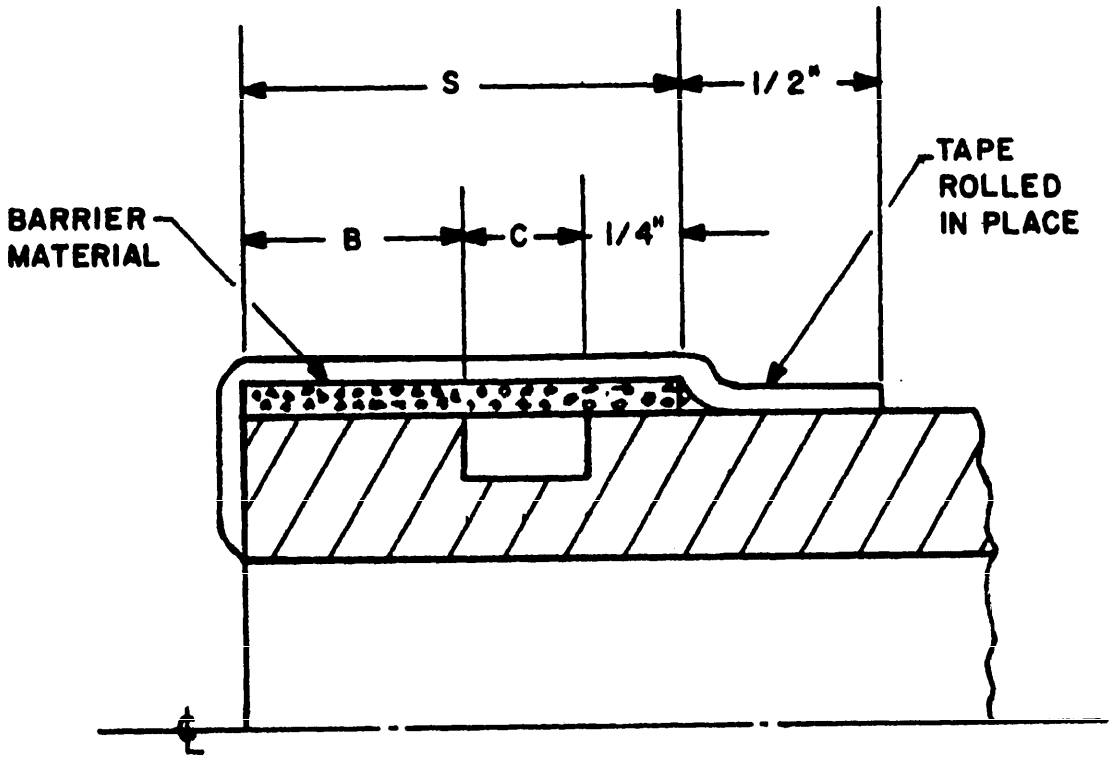
DETAIL A

NOTE:
All Dimensions In Inches.

FIGURE 1. Aluminum alloy tubing.

X-3896

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S ——— LAND SEALING SURFACE

B ——— LAND

C ——— GROOVE

FIGURE 2. Land sealing surface for grooved end pipe, tube end fittings.

X-449A

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-10794G Tubes, Aluminum-Alloy, Extruded Pipeline Section, With Grooved Nipple Welded on Each End

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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